

1D CNN — Architecture Variant Comparison

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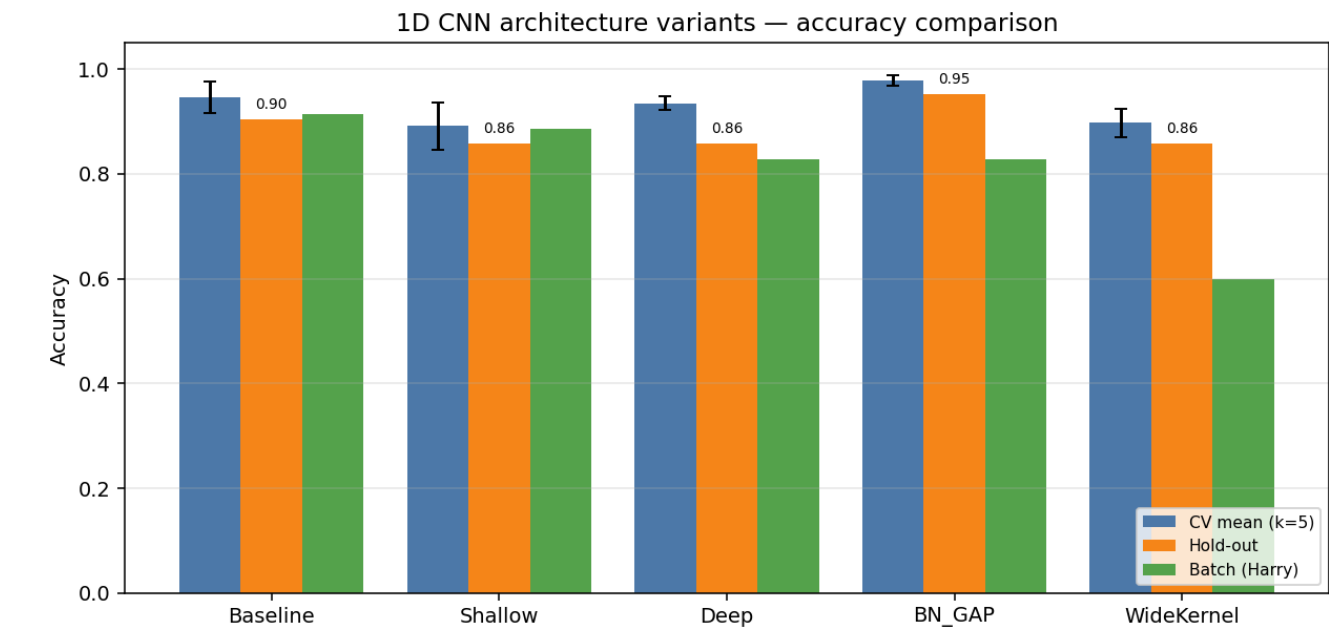
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Total trials	206 (train pool 185, hold-out 21)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

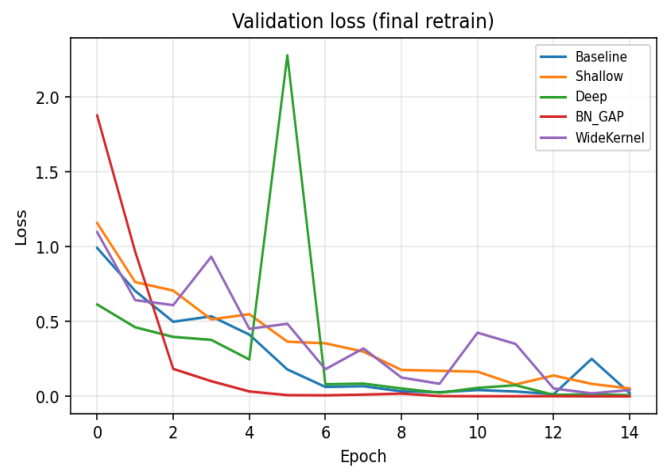
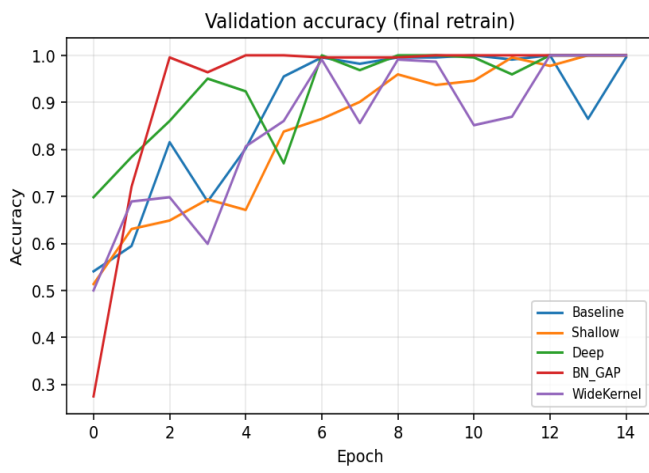
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9459	0.0296	0.9048	0.8533	0.9143	32/35
Shallow	72,487	0.8919	0.0452	0.8571	1.1375	0.8857	31/35
Deep	204,007	0.9351	0.0132	0.8571	1.0952	0.8286	29/35
BN_GAP	57,319	0.9784	0.0108	0.9524	0.1530	0.8286	29/35
WideKernel	131,687	0.8973	0.0265	0.8571	1.3242	0.6000	21/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9459 ± 0.0296
Hold-out test acc / loss	0.9048 · 0.8533
Batch-test acc	0.9143 (32/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	888	37	0.9459	0.9730	0.9459
2	888	37	0.9730	0.9730	0.9730
3	888	37	0.8919	0.8919	0.8919
4	888	37	0.9459	0.9730	0.9459
5	888	37	0.9730	0.9730	0.9730

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.667	0.800	3
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	0.667	0.667	0.667	3
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	1.000	1.000	1.000	2
macro avg	0.924	0.905	0.908	21
weighted avg	0.914	0.905	0.903	21

Batch test per class (Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	35	0.914

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8919 ± 0.0452
Hold-out test acc / loss	0.8571 · 1.1375
Batch-test acc	0.8857 (31/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	888	37	0.9459	0.9459	0.9459
2	888	37	0.8108	0.8108	0.8108
3	888	37	0.9189	0.9189	0.9189
4	888	37	0.8919	0.8919	0.8919
5	888	37	0.8919	0.8919	0.8919

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.750	1.000	0.857	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	0.750	1.000	0.857	3
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	0.500	0.667	2
macro avg	0.893	0.833	0.820	21
weighted avg	0.893	0.857	0.836	21

Batch test per class (Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	31	35	0.886

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9351 ± 0.0132
Hold-out test acc / loss	0.8571 · 1.0952
Batch-test acc	0.8286 (29/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	888	37	0.9459	1.0000	0.9459
2	888	37	0.9459	0.9730	0.9459
3	888	37	0.9189	0.9189	0.9189
4	888	37	0.9459	0.9730	0.9459
5	888	37	0.9189	0.9730	0.9189

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	0.667	0.667	0.667	3
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	0.667	1.000	0.800	4
Drum_Roll	1.000	1.000	1.000	2
macro avg	0.905	0.857	0.852	21
weighted avg	0.889	0.857	0.843	21

Batch test per class (Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	0	5	0.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	29	35	0.829

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9784 ± 0.0108
Hold-out test acc / loss	0.9524 · 0.1530
Batch-test acc	0.8286 (29/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	888	37	0.9730	0.9730	0.9730
2	888	37	0.9730	1.0000	0.9730
3	888	37	0.9730	0.9730	0.9730
4	888	37	1.0000	1.0000	1.0000
5	888	37	0.9730	0.9730	0.9730

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.667	0.800	3
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	1.000	1.000	1.000	3
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	1.000	1.000	1.000	2
macro avg	0.971	0.952	0.956	21
weighted avg	0.962	0.952	0.950	21

Batch test per class (Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	1	5	0.200
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	29	35	0.829

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8973 ± 0.0265
Hold-out test acc / loss	0.8571 · 1.3242
Batch-test acc	0.6000 (21/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	888	37	0.8919	0.8919	0.8919
2	888	37	0.8919	0.8919	0.8919
3	888	37	0.9459	0.9459	0.9459
4	888	37	0.8919	0.9730	0.8919
5	888	37	0.8649	0.9189	0.8649

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	0.667	0.667	0.667	3
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	0.667	1.000	0.800	4
Drum_Roll	1.000	1.000	1.000	2
macro avg	0.905	0.857	0.852	21
weighted avg	0.889	0.857	0.843	21

Batch test per class (Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	2	5	0.400
Double_Raise	0	5	0.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	4	5	0.800
Overall	21	35	0.600